



Paper Airplane Design Process Worksheet

IED Lesson 0.4 - Engineering Design Process | Small-team distance challenge | Individual worksheet

Name:		Period:		Team:	
Date:		Document #1		Notebook:	Add to TOC

Challenge: Design, build, test, improve, and retest a paper airplane for distance. Your team will build one Version 1 prototype and one revised Version 2 prototype. Each version gets one official test flight.

Material constraint: one sheet of paper for Version 1 and one new sheet of paper for Version 2 - nothing else. No tape, scissors, paper clips, added weight, or extra materials.

Measurement rule: measure from the launch line to where the airplane first touches the ground. Record distance in feet and inches.

1. Define Problem

Write the problem your team is trying to solve. Include the goal, constraints, and how success will be measured.

2. Generate Concepts

Sketch your own paper airplane concept. Label at least three features that you think could help flight distance.

Which team concept did your group choose to build for Version 1, and why?



3. Develop Solution

Before building, describe your team plan for Version 1. What features are you using, and why do you think they will help distance?

4. Construct and Test

Version	Design Change	Distance	Notes
Version 1	Initial prototype	___ ft ___ in	
Version 2	Revision after testing	___ ft ___ in	

Explain what your team changed from Version 1 to Version 2 and why.

5. Evaluate Solution

Did Version 2 improve, get worse, or stay about the same compared with Version 1? Use distance evidence to support your answer.

Which design process step was most important during this challenge? Explain why.

6. Present Solution

If your team had to briefly present your design, what would you say was your best design decision and your next improvement?

Grading focus: completion, clear evidence, and thoughtful responses - not how far your airplane flies.